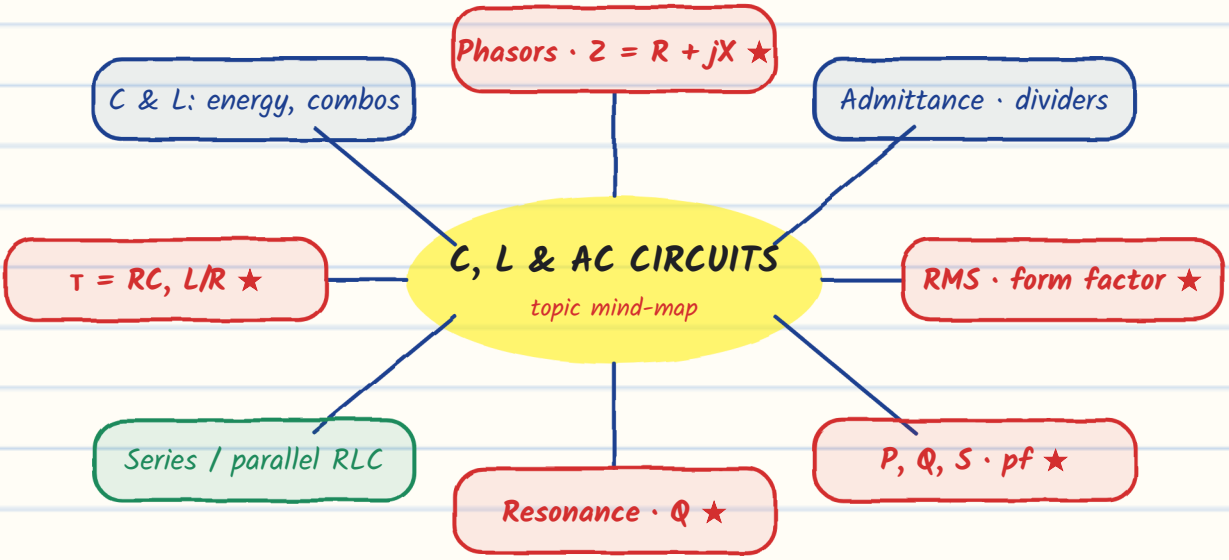
Capacitors, Inductors & AC Circuits

C & L · transients · RMS · phasors · impedance · resonance · AC power: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

➜ Syllabus coverage

"Capacitors and inductors: inductors and capacitors, their characteristics, series-parallel combination of inductors and capacitors; RLC transients. Series and parallel AC circuits: impedance and phasor diagram, series and parallel networks, voltage divider rule, admittance and susceptance". Sinusoid values, resonance and AC power are added because they are tested with this topic. AC mesh/nodal analysis, Y–Δ and the theorems reuse Lesson 2's methods with Z in place of R.

High-yield list

① ½CV², ½LI²
② τ = RC, L/R (63.2%)
③ 0.707 V_m, FF 1.11
④ ELI the ICE man
⑤ $f_0 = 1/(2\pi\sqrt{LC})$
⑥ P = VI cos φ

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

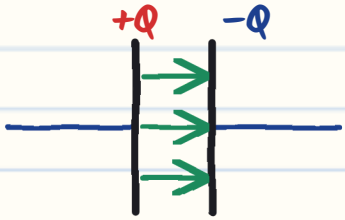
➜ Contents

✓ Capacitor & inductor, combinations … p. 2
✓ RC / RL transients, RLC damping … p. 3
✓ Sinusoids: RMS, average, FF ..... p. 4
✓ Phasors & complex numbers ..... p. 5
✓ Impedance, admittance, dividers … p. 6
✓ Series & parallel RLC ..... p. 7
✓ Resonance, Q, bandwidth ..... p. 8
✓ AC power & power factor ..... p. 9
✓ Night-before sheet ..... p. 10
✓ Practice MCQs + written ..... p. 11

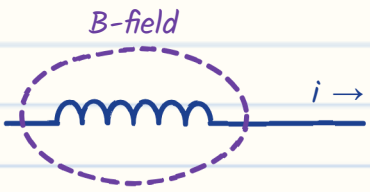
1 Capacitor & Inductor: the Storage Elements high-yield

Capacitor Two conductors separated by a dielectric. Stores energy in the **electric field**. $C = Q/V$, unit **farad** ($1\ F = 1\ C/V$). Parallel plates: $C = \varepsilon A/d$.

Inductor A coil of wire. Stores energy in the **magnetic field**. $L = N\Phi/I$, unit **henry** ($1\ H = 1\ Wb/A = 1\ V{\cdot}s/A$). Solenoid: $L = \mu N^2 A/l$.



E-field: $W = \tfrac{1}{2}CV^2$



B-field: $W = \tfrac{1}{2}LI^2$

C keeps charge on its plates; L keeps flux in its core

Property	Capacitor C	Inductor L
Defining law	$i = C\,dv/dt$	$v = L\,di/dt$
Integral form	$v = (1/C)\int i\,dt$	$i = (1/L)\int v\,dt$
Energy stored	$\tfrac{1}{2}CV^2 = Q^2/2C = \tfrac{1}{2}QV$	$\tfrac{1}{2}LI^2$
Cannot change instantly	**voltage**	**current**
DC steady state	open circuit	short circuit
At switch-on (no initial energy)	short circuit	open circuit
AC phase	I **leads** V by 90°	I **lags** V by 90°
Series	$1/C = \Sigma\, 1/C_k$	$L = \Sigma\, L_k$
Parallel	$C = \Sigma\, C_k$	$1/L = \Sigma\, 1/L_k$

Memory trick **L behaves like R** (series adds). **C is R upside-down** (parallel adds). C and L are **duals**: swap $v \leftrightarrow i$ and $C \leftrightarrow L$ to turn one row into the other.

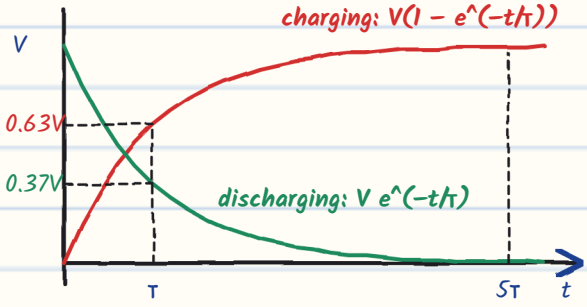
Worked example ✍️ 2 µF, 3 µF, 6 µF on 100 V. **Series:** $1/C = \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{6} = 1 \rightarrow C = 1\ \mu F$; $Q = CV = 100\ \mu C$ on **each**; $V = Q/C \rightarrow 50$ V, 33.3 V, 16.7 V (sum 100 ✓); $W = \tfrac{1}{2} \times 1\,\mu \times 100^2 =$ **5 mJ**. **Parallel:** $C = 11\ \mu F$, $W = 55$ mJ.
Inductors 4 H ‖ 12 H (no mutual coupling) = 48/16 = **3 H**; at 2 A, $W = \tfrac{1}{2} \times 3 \times 2^2 =$ **6 J**.

Exam trap Capacitors in series carry the **same charge**, so the **smallest C takes the largest voltage** (2 µF gets 50 V). Series C is always smaller than the smallest one. Ideal C and L dissipate **no power**: they store energy and give it back.

2 RC & RL Transients (First Order) written favourite

Transient The change from one steady state to another after a switch operates. In a circuit with one C or one L it is an **exponential** set by the time constant τ: **$\tau = RC$** or **$\tau = L/R$** (R = Thevenin resistance seen by C or L).

One formula for every case: $x(t) = x(\infty) + [x(0^+) - x(\infty)]\, e^{-t/\tau}$ ($x = v_C$ or i_L)



t	charging	decay left
1τ	**63.2%**	**36.8%**
2τ	86.5%	13.5%
3τ	95.0%	5.0%
4τ	98.2%	1.8%
5τ	99.3%	0.7%

RC charge: $v_C = V(1 - e^{-t/RC})$

RL rise: $i = (V/R)(1 - e^{-Rt/L})$

RC discharge: $v_C = V_0 e^{-t/RC}$

RL decay: $i = I_0 e^{-Rt/L}$

Worked example: RC ✍ 10 kΩ, 100 μF, 12 V: τ = **1 s**; $i(0^+)$ = 1.2 mA; v_C(1 s) = 12 × 0.632 = **7.59 V**; 6 V at $t = \tau \ln 2$ = **0.693 s**

Worked example: RL ✍ 2 H, 4 Ω, 12 V: $\tau = L/R$ = **0.5 s**; $i(\infty)$ = 3 A; i(0.5 s) = 3 × 0.632 = **1.90 A**; final energy ½ × 2 × 3² = **9 J**

Memory trick 63 – 86 – 95 – 98 – 99: one number per τ. "**Five taus and it's done.**"

Exam trap RL: $\tau = L/R$, not LR. A bigger R makes RL **faster** but RC **slower**. i_C and v_L decay ($e^{-t/\tau}$).

3 Second-Order (RLC) Response

$\omega_0 = 1/\sqrt{LC}$ (undamped natural frequency) · damping factor $\alpha = R/2L$ (series), $\alpha = 1/(2RC)$ (parallel) · $\zeta = \alpha/\omega_0$ · damped frequency $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

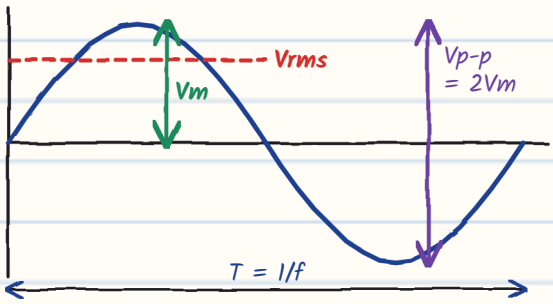
Condition	Response	Series RLC
$\alpha > \omega_0$ ($\zeta > 1$)	overdamped: two exponentials, slow	$R > 2\sqrt{L/C}$
$\alpha = \omega_0$ ($\zeta = 1$)	critically damped: **fastest** without oscillation	$R = 2\sqrt{L/C}$
$\alpha < \omega_0$ ($\zeta < 1$)	underdamped: decaying oscillation at ω_d	$R < 2\sqrt{L/C}$

e.g. L = 1 H, C = 1 μF → ω_0 = 1000 rad/s; critical R (series) = $2\sqrt{10^6}$ = **2 kΩ**.

4 Sinusoids: Peak, RMS, Average

high-yield

Sinusoid $v(t) = V_m \sin(\omega t + \varphi)$: V_m = amplitude (peak), $\omega = 2\pi f$ rad/s, $T = 1/f$, φ = phase angle. Bangladesh mains: **220 V rms, 50 Hz**.



RMS The DC value that gives the **same heating** in a resistor: $V_{rms} = \sqrt{(\text{mean of } v^2)}$. Meters and ratings ("220 V") are RMS.

Average Mean over a **half cycle**: $2V_m/\pi$. Over a full cycle a pure sine averages **0**.

Waveform (peak V_m)	V_{rms}	V_{avg}	FF = rms/avg	Crest = peak/rms
Sine / full-wave rectified	**0.707 V_m**	**0.637 V_m**	**1.11**	**1.414**
Half-wave rectified sine	$V_m/2$	V_m/π	1.57	2
Square ($\pm V_m$)	V_m	V_m	1	1
Triangle / sawtooth	$V_m/\sqrt{3}$	$V_m/2$	1.155	$\sqrt{3} = 1.732$

Worked example 1 BD mains 220 V, 50 Hz: $V_m = 220\sqrt{2} =$ **311 V**; $V_{p\text{-}p} =$ 622 V; $\omega = 2\pi \times 50 =$ **314 rad/s**; $T =$ **20 ms**; $V_{avg} = 0.637 \times 311 = 198$ V.

Worked example 2 $v = 100 \sin(314t + 30°)$ V: $V_m = 100$ V, $V_{rms} =$ **70.7 V**, $f = 314/2\pi \approx$ **50 Hz**, $\varphi = +30°$; at $t = 0$, $v = 100 \sin 30° =$ **50 V**.

→ Phase difference

$v = 10 \sin(\omega t + 20°)$, $i = 2 \sin(\omega t - 40°)$ → v **leads** i by $20° - (-40°) = 60°$. Write both as sin with positive amplitude first: $\cos \omega t = \sin(\omega t + 90°)$, $-\sin \omega t = \sin(\omega t + 180°)$.

Derivation (written) $V_{rms}^2 = (1/T)\int_0^T V_m^2 \sin^2\omega t\, dt = (V_m^2/T)\int_0^T \tfrac{1}{2}(1 - \cos 2\omega t)\, dt = V_m^2/2 \rightarrow$ **$V_{rms} = V_m/\sqrt{2}$**. Half-cycle average: $(1/\pi)\int_0^\pi V_m \sin\theta\, d\theta = 2V_m/\pi$.

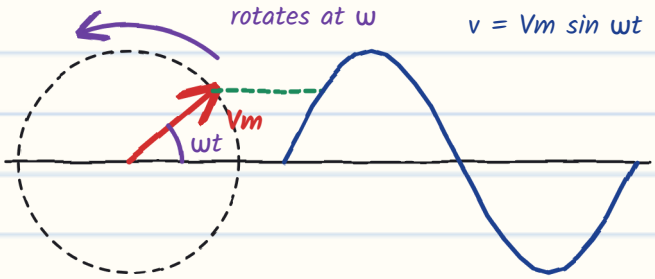
Memory trick The sine's four numbers: **0.707 · 0.637 · 1.11 · 1.414** (rms, average, form, crest).

Exam trap "220 V" is RMS; insulation must withstand the **peak 311 V**. Full-wave rectified sine has the **same** rms and average as a sine.

5 Phasors & Complex Numbers

high-yield

Phasor A complex number that carries the magnitude and phase of a sinusoid of a known frequency: $v = V_m \sin(\omega t + \varphi) \rightarrow \mathbf{V} = V\angle\varphi$ (V usually rms). Think of a vector rotating anticlockwise at ω; its vertical projection traces the sine. It turns calculus ($d/dt \rightarrow j\omega$) into algebra.



rotating phasor and the sine it projects

Rectangular ↔ polar

$a + jb = r\angle\theta$, $r = \sqrt{(a^2 + b^2)}$, $\theta = \tan^{-1}(b/a)$

$a = r\cos\theta$, $b = r\sin\theta$

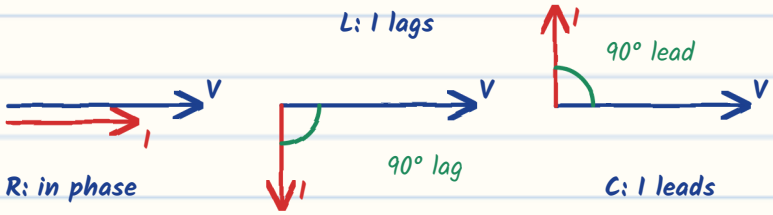
$j = \sqrt{-1} = 1\angle 90°$, $j^2 = -1$, $1/j = -j$

Add/subtract in **rectangular**; multiply/divide in **polar** (multiply r, add angles)

Worked example 1 $I = 100\angle 0° / (3 + j4) = 100\angle 0° / 5\angle 53.13° = 20\angle -53.13°$ A $= 12 - j16$ A

Worked example 2 $10\angle 0° + 10\angle 90° = 10 + j10 = 14.14\angle 45°$, not 20

Element	Time domain	Phasor law	Impedance Z	Phase of I w.r.t. V
R	$v = Ri$	$V = RI$	R	in phase (0°)
L	$v = L\,di/dt$	$V = j\omega L\, I$	$jX_L = j\omega L$	I **lags** by 90°
C	$i = C\,dv/dt$	$V = I/(j\omega C)$	$-jX_C = 1/(j\omega C)$	I **leads** by 90°



phasor diagrams with V as reference (anticlockwise = leading)

Memory trick **ELI the ICE man**: in L, E (voltage) comes before I; in C, I comes before E. Or **CIVIL**: in C, I before V; V before I in L.

Exam trap Phasors only combine sinusoids of the **same frequency**. Adding peak values of out-of-phase waves (10 + 10 = 20) is wrong: add them as complex numbers.

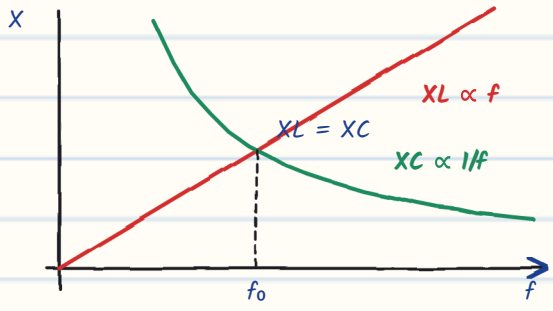
6 Impedance, Admittance & AC Networks

MCQ + Written

Impedance $Z = V/I = R + jX = |Z| \angle\theta$ (Ω). R = resistance, X = reactance. $|Z| = \sqrt{(R^2 + X^2)}$, $\theta = \tan^{-1}(X/R)$. $X > 0$ inductive, $X < 0$ capacitive.

Admittance $Y = 1/Z = G + jB$, unit **siemens (S)**. G = conductance, B = susceptance. $G = R/(R^2 + X^2)$, $B = -X/(R^2 + X^2)$.

	Inductor	Capacitor
Reactance	$X_L = 2\pi fL$	$X_C = 1/(2\pi fC)$
In Z	$+jX_L$	$-jX_C$
Susceptance	$B_L = -1/(\omega L)$	$B_C = +\omega C$
$f \to 0$ (DC)	$X_L = 0$: short	$X_C \to \infty$: open
$f \to \infty$	open	short
f doubled	X_L doubles	X_C halves



reactance against frequency

Network rules (same as DC, with Z) Series: $Z = Z_1 + Z_2 + \ldots$ · Parallel: $Y = Y_1 + Y_2 + \ldots$, two branches $Z = Z_1Z_2/(Z_1 + Z_2)$

Voltage divider: $V_1 = V \cdot Z_1/(Z_1 + Z_2)$ · Current divider: $I_1 = I \cdot Z_2/(Z_1 + Z_2) = I \cdot Y_1/(Y_1 + Y_2)$

Worked example 1: reactance ✍ 0.1 H at 50 Hz: $X_L = 2\pi \times 50 \times 0.1 =$ **31.4 Ω**. 100 µF at 50 Hz: $X_C = 1/(2\pi \times 50 \times 10^{-4}) =$ **31.8 Ω**.

Worked example 2: parallel ✍ $R = 10\ \Omega \parallel X_C = 10\ \Omega$ on $100\angle 0°$ V. $Y = 0.1 + j0.1 =$ **$0.1414\angle 45°$ S**; $Z = 1/Y = 7.07\angle -45° =$ **$5 - j5\ \Omega$**; $I = VY =$ **$14.14\angle 45°$ A** (leading).

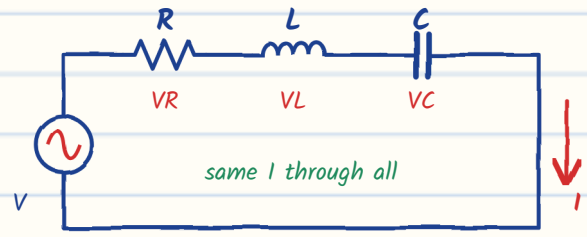
Worked example 3: divider ✍ $100\angle 0°$ V across $R = 6\ \Omega$ in series with $jX_L = j8\ \Omega$. $Z = 10\angle 53.13°$. $V_L = 100 \times j8/(6 + j8) =$ **$80\angle 36.87°$ V**, $V_R =$ **$60\angle -53.13°$ V**. Check: $60 + 80 = 140 \neq 100$, but $\sqrt{(60^2 + 80^2)} =$ **100** ✓.

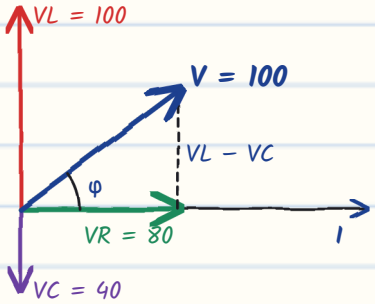
Exam trap $G = 1/R$ **only** if the branch has no reactance; for $R + jX$, $G = R/(R^2 + X^2)$. Voltage magnitudes in AC **do not add** arithmetically.

Lesson 2 link AC mesh and nodal analysis, Thevenin, Norton and superposition work exactly as in Lesson 2 (Nodal, Mesh & Circuit Theorems), using complex Z. Balanced $\Delta \to Y$: $Z_Y = Z_\Delta/3$. AC maximum power transfer: $Z_L = Z_{th}^*$ (complex conjugate), $P_{max} = |V_{th}|^2/(4R_{th})$.

7 Series RLC Circuit

written favourite





Formulas $Z = R + j(X_L - X_C)$ · $|Z| = \sqrt{(R^2 + (X_L - X_C)^2)}$ · $\varphi = \tan^{-1}((X_L - X_C)/R)$ · $V = \sqrt{(V_R^2 + (V_L - V_C)^2)}$ · $pf = \cos\varphi = R/|Z|$

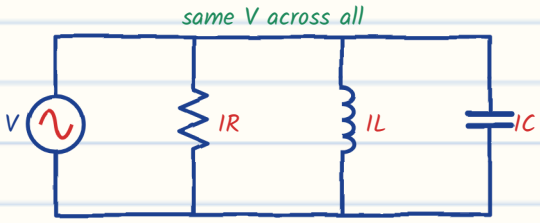
$X_L > X_C$: **inductive**, I lags · $X_L < X_C$: **capacitive**, I leads · $X_L = X_C$: **resonance**, $Z = R$

Worked example ✍️ $R = 8\ \Omega$, $X_L = 10\ \Omega$, $X_C = 4\ \Omega$ on 100 V rms.

1. $X = 10 - 4 = 6\ \Omega$ (inductive); $Z = \sqrt{(8^2 + 6^2)} = 10\ \Omega$; $\varphi = \tan^{-1}(6/8) = 36.87°$
2. $I = 100/10 = \mathbf{10\ A}$, lagging; $V_R = 80$ V, $V_L = 100$ V, $V_C = 40$ V; check $\sqrt{(80^2 + 60^2)} = 100$ ✓
3. $pf = 8/10 = \mathbf{0.8\ lagging}$; $P = I^2R = \mathbf{800\ W}$, $Q = I^2X = 600$ VAR, $S = VI = 1000$ VA

Exam trap In series RLC the voltage across L or C can **equal or exceed the supply** (here $V_L = 100\ V = V$). Only the phasor sum must equal V.

8 Parallel RLC Circuit



$Y = 1/R + j(\omega C - 1/\omega L)$ · $I = V \cdot Y$

$I = \sqrt{(I_R^2 + (I_C - I_L)^2)}$

$I_L > I_C$ ($X_L < X_C$): inductive, lagging

$I_C > I_L$: capacitive, leading

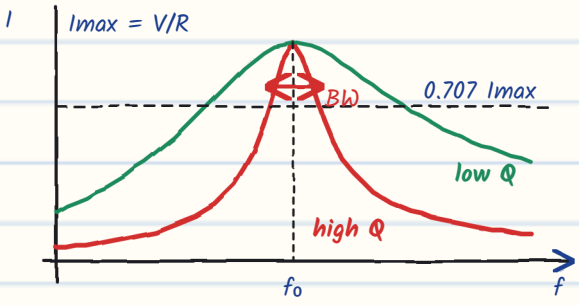
Worked example ✍️ $R = 20\ \Omega$, $X_L = 10\ \Omega$, $X_C = 20\ \Omega$ in parallel on 100 V. $I_R = 5$ A, $I_L = 10$ A $(-j)$, $I_C = 5$ A $(+j)$. $I = 5 - j10 + j5 = 5 - j5 = 7.07\angle -45°$ A; $Z = 100/7.07 = \mathbf{14.14\ \Omega}$; $pf = 0.707$ lagging.

Memory trick **Series: the bigger X wins. Parallel: the smaller X wins** (it draws the bigger current). Same $X_L < X_C$ → capacitive in series, inductive in parallel.

9 Resonance, Q-factor & Bandwidth

high-yield

Resonance The frequency at which $X_L = X_C$, so the circuit is purely resistive (pf = 1, V and I in phase): $\omega_0 = 1/\sqrt{(LC)}$, $f_0 = 1/(2\pi\sqrt{(LC)})$. f_0 does not depend on R.



series resonance curve (half-power line dashed)

Quality factor (series): $Q = \omega_0 L/R = 1/(\omega_0 CR) = (1/R)\sqrt{(L/C)}$

Bandwidth: $BW = f_0/Q = R/(2\pi L)$ Hz

Half-power (−3 dB) points f_1, f_2: $I = I_{max}/\sqrt{2}$, $P = P_{max}/2$

$f_0 = \sqrt{(f_1 f_2)}$; for high Q, $f_{1,2} \approx f_0 \mp BW/2$

High Q → sharp peak, **better selectivity**

At resonance	Series RLC	Parallel (tank)
Impedance	**minimum**, $Z = R$	**maximum** (ideal: R; coil r: L/Cr)
Line current	**maximum**, V/R	**minimum**
Magnification	voltage: $V_L = V_C = QV$	current: $I_L = I_C = QI$
Q	$\omega_0 L/R$	$R/(\omega_0 L) = \omega_0 CR$
Name	acceptor circuit	rejector circuit
Below / above f_0	capacitive / inductive	inductive / capacitive

Practical tank (coil r + L, with C across): $f_0 = (1/2\pi)\sqrt{(1/(LC) - r^2/L^2)}$, slightly below $1/(2\pi\sqrt{LC})$; dynamic impedance $Z_D = L/(Cr)$.

Worked example ✍️ Series RLC: L = 10 mH, C = 1 µF, R = 10 Ω, V = 10 V.

1. $\omega_0 = 1/\sqrt{(10^{-2} \times 10^{-6})} = 10^4$ **rad/s**; $f_0 = 10^4/2\pi =$ **1.59 kHz** (not 10 kHz)
2. $X_L = X_C = \omega_0 L = 100\ \Omega$; $I_0 = V/R =$ **1 A**
3. $Q = 100/10 =$ **10**; $BW = 1591/10 =$ **159 Hz**; $V_L = V_C = Q \cdot V =$ **100 V** on a 10 V supply

Memory trick **Series = Small Z, Sucks current in** (acceptor). **Parallel = Peak Z, Pushes current away** (rejector).

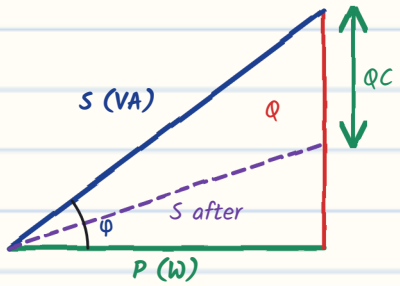
Exam trap $f_0 \propto 1/\sqrt{(LC)}$: C × 4 → f_0 **halves**. Adding R changes Q and BW, never f_0.

IT link Radio, Wi-Fi and mobile receivers tune by resonance; band-pass filters; quartz crystals (a very high-Q resonator) set CPU and RTC clocks; RFID/NFC tags resonate at 13.56 MHz.

10 AC Power & Power Factor high-yield

With $v = V_m \sin \omega t$ and $i = I_m \sin(\omega t - \varphi)$: $p = vi = \mathbf{VI \cos \varphi - VI \cos(2\omega t - \varphi)}$ (rms V, I). The second term averages to zero (it pulses at **twice** the supply frequency), so the average power is VI cos φ.

Power	Formula	Unit
Real / active P	VI cos φ = I^2R	W
Reactive Q	VI sin φ = I^2X	VAR
Apparent S	VI = $I^2\lvert Z\rvert$	VA
Complex S	$V \cdot I^* = P + jQ$	VA
Triangle	$S^2 = P^2 + Q^2$	



power triangle; a capacitor cuts Q

Power factor $pf = \cos \varphi = P/S = R/\lvert Z\rvert$, between 0 and 1. **Lagging**: inductive load (motors, fans, transformers, ballasts), I lags V. **Leading**: capacitive load. Pure R: pf = 1, Q = 0. Pure L or C: pf = 0, P = 0. Inductive Q is positive (absorbed), capacitive Q negative.

Why low pf is bad For the same P, $I = P/(V \cos \varphi)$ is larger → more I^2R line loss, thicker cables, bigger transformers, worse voltage drop. Utilities penalise industrial consumers with low pf.

Memory trick **P does the work, Q sloshes back and forth, S is what the wires carry.** Units: **W**atts work, **VAR** varies, **VA** is the total.

pf correction Connect a capacitor in parallel: $Q_C = P(\tan \varphi_1 - \tan \varphi_2)$, $C = Q_C/(\omega V^2)$. P is unchanged; S and line current fall.

Worked example ✍ 10 kW load, pf 0.8 lagging, 220 V, 50 Hz; raise pf to 0.95.

1. $\tan \varphi_1 = 0.75$ ($Q_1 = 7.5$ kVAR, $S_1 = 12.5$ kVA); $\tan \varphi_2 = \tan(\cos^{-1} 0.95) = 0.329$
2. $Q_C = 10\,000 \times (0.75 - 0.329) = \mathbf{4.21\ kVAR}$; $C = 4213/(314.16 \times 220^2) =$ **277 µF**
3. Line current $10\,000/(220 \times 0.8) = 56.8$ A → $10\,000/(220 \times 0.95) = \mathbf{47.8\ A}$ (−16%)

Exam trap A wattmeter reads **P**, not VI. Generators, transformers and UPSs are rated in **VA** because their heating depends on current, whatever the pf.

IT link A 1000 VA UPS at pf 0.6 supplies only **600 W**; server PSUs use active PFC (pf > 0.9). 1 unit = **1 kWh** = 3.6 MJ.

Night-Before-the-Exam Sheet

Formula sheet

$i = C\,dv/dt$ · $W = \tfrac{1}{2}CV^2$

$v = L\,di/dt$ · $W = \tfrac{1}{2}LI^2$

C series: $1/C = \Sigma 1/C_k$ · parallel: ΣC_k

L series: ΣL_k · parallel: $1/L = \Sigma 1/L_k$

$\tau = RC$ · $\tau = L/R$ · $1\tau \rightarrow 63.2\%$

$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-t/\tau}$

$V_{rms} = 0.707V_m$ · $V_{avg} = 0.637V_m$

FF = 1.11 · crest = 1.414 (sine)

$X_L = 2\pi fL$ · $X_C = 1/(2\pi fC)$

$Z = R + j(X_L - X_C)$ · $Y = G + jB$

$f_0 = 1/(2\pi\sqrt{LC})$ · $Q = \omega_0 L/R$

$BW = f_0/Q$ · $f_0 = \sqrt{(f_1 f_2)}$

$P = VI\cos\varphi$ · $Q = VI\sin\varphi$ · $S = VI$

$S^2 = P^2 + Q^2$ · $pf = P/S = R/|Z|$

$Q_C = P(\tan\varphi_1 - \tan\varphi_2)$

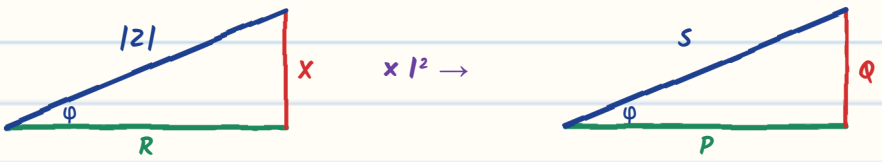
series $\alpha = R/2L$ · parallel $\alpha = 1/2RC$

→ Top 12 one-liners for MCQs

1. C blocks DC; L passes DC (short)
2. v_C and i_L cannot jump
3. BD mains: 220 V rms, 50 Hz, 311 V peak
4. Pure L: I lags 90° · pure C: I leads 90°
5. Unit of Y, G, B: siemens
6. Unit of Q: VAR · unit of S: VA
7. Series resonance: Z min, I max (acceptor)
8. Parallel resonance: Z max, I min (rejector)
9. At resonance pf = 1
10. Q ↑ → BW ↓ → selectivity ↑
11. Critical damping: $R = 2\sqrt{(L/C)}$ (series)
12. Low pf → more current, more loss

Top traps $\tau = L/R$ not LR · series C carry equal charge (smallest C gets most V) · "220 V" is rms · AC voltages add as phasors · $G \neq 1/R$ in a series R-X branch · ω_0 in rad/s ≠ f_0 in Hz · series: bigger X wins, parallel: smaller X wins · f_0 does not depend on R · wattmeter reads P

All tricks together L adds like R, C is R upside-down · 63-86-95-98-99, five taus and done · sine numbers 0.707, 0.637, 1.11, 1.414 · ELI the ICE man / CIVIL · Series Small Z (acceptor), Parallel Peak Z (rejector) · P works, Q sloshes, S is carried



impedance triangle × I^2 = power triangle (same angle φ)

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Energy stored in a 10 µF capacitor charged to 100 V:
(a) 1 mJ (b) 0.05 J
(c) 0.5 J (d) 50 J

2. In DC steady state an ideal inductor acts as:
(a) a short circuit (b) an open circuit
(c) a resistor (d) a voltage source

3. Time constant of R = 1 MΩ, C = 2 µF:
(a) 0.5 s (b) 1 s
(c) 2 s (d) 20 s

4. Form factor of a sine wave:
(a) 1.414 (b) 1.0
(c) 1.57 (d) 1.11

5. In a purely capacitive circuit, the current:
(a) lags V by 90° (b) is in phase
(c) leads V by 90° (d) lags V by 180°

6. At series resonance the impedance is:
(a) minimum, = R (b) maximum
(c) zero (d) infinite

7. Power factor of a series circuit with R = 6 Ω, X = 8 Ω:
(a) 0.6 (b) 0.8
(c) 0.75 (d) 1

8. Unit of reactive power:
(a) W (b) VA
(c) VAR (d) J

9. In a series LC circuit C is made 4 times larger. f_0 becomes:
(a) $4f_0$ (b) $2f_0$
(c) $f_0/4$ (d) $f_0/2$

10. The unit of admittance is:
(a) ohm (b) siemens
(c) henry (d) farad

Answer key → 1-b · 2-a · 3-c · 4-d · 5-c · 6-a · 7-a · 8-c · 9-d · 10-b

Written question $v = 141.4 \sin(1000t)$ V is applied to a series circuit of R = 30 Ω, L = 80 mH and C = 25 µF. Find X_L, X_C, Z, I, the power factor, P, Q, S, and the resonant frequency. Draw the phasor diagram. 10 marks

1. $V = 141.4/\sqrt{2} = 100$ V rms, $\omega = 1000$ rad/s; $X_L = \omega L = 80\ \Omega$, $X_C = 1/(\omega C) = 40\ \Omega$ — 2
2. $Z = 30 + j40 = 50\angle 53.13°\ \Omega$ (inductive); $I = 100/50 = 2$ A, lagging 53.13° — 2
3. pf = 30/50 = 0.6 lag; $P = I^2R = 120$ W, $Q = I^2X = 160$ VAR, $S = VI = 200$ VA ✓ — 3
4. $V_R = 60$ V, $V_L = 160$ V, $V_C = 80$ V; phasor diagram as on p. 7 — 1
5. $\omega_0 = 1/\sqrt{(0.08 \times 25\ \mu)} = 707.1$ rad/s → $f_0 = 112.5$ Hz < f = 159.2 Hz, so inductive ✓ — 2

→ Before the exam ✓

- ✓ Can write the C/L duality table
- ✓ Can solve an RC/RL switching problem
- ✓ Know rms, average, FF, crest of 4 waves
- ✓ Can draw series RLC phasor diagram
- ✓ Can find f_0, Q, BW and size a pf capacitor

Also practise (written):
① Derive $W = \frac{1}{2}CV^2$
② Derive RC charging
③ Prove $V_{rms} = V_m/\sqrt{2}$
④ Series vs parallel resonance